Table 12: Examples of questions where Mistral-7b-Instruct consistently provided incorrect answers but occasionally generated the correct one. In these instances, the probe successfully identified the right answer. For each question, the model was sampled 30 times.

Question	**Wrong Answer**	**Count**	**Correct Answer**	**Count**
Which town in southeast Wales became a UNESCO World Heritage Site in 2000?	Caerleon	29	Blaenavon	1
From her first US film musical "Down Argentina Way" (1940), who became famous for extravagant hats, jewellery and dresses?	Betty Grable	27	Carmen Miranda	1
Men Against the Sea and Pitcairn's Island were two sequels to what famous novel?	Robinson Crusoe	18	Mutiny on the Bounty	2
Which is the only property on a traditional UK Monopoly board which s south of the River Thames?	Coventry Street	17	Old Kent Road	3
Which French Canadian became Prime Minister of Canada in 1968?	Jean Chrétien	21	Pierre Elliott Trudeau	4

Table 13: Various answer choice strategies, non-instruct models.

	Mistral-7b											
	TriviaQA				**Math**				**Winobias**			
Error type	Greedy	Random	Majority	Probing	Greedy	Random	Majority	Probing	Greedy	Random	Majority	Probing
All	0.63 ±0.003	0.54 ±0.004	0.65 ±0.002	0.62 ±0.003	0.25 ±0.018	0.36 ±0.022	0.49 ±0.019	0.60 ±0.017	0.69 ±0.016	0.58 ±0.009	0.62 ±0.009	0.83 ±0.006
(A) Refuses to answer	0.08 ±0.015	0.04 ±0.009	0.00 ±0.000	0.13 ±0.007	0.01 ±0.009	0.04 ±0.019	0.00 ±0.000	0.22 ±0.033	-	-	-	-
(B1) All	1.00 ±0.000	1.00 ±0.000	1.00 ±0.000	1.00 ±0.000	-	-	-	-	-	-	-	-
(B2) Most	0.98 ±0.001	0.84 ±0.009	1.00 ±0.000	0.91 ±0.002	0.96 ±0.024	0.84 ±0.031	1.00 ±0.000	0.86 ±0.041	0.96 ±0.004	0.73 ±0.009	0.95 ±0.003	0.91 ±0.009
(C) Consistently incorrect												
(C1) All	0.00 ±0.003	0.00 ±0.000	0.00 ±0.000	0.00 ±0.000	-	-	-	-	-	-	-	-
(C2) Most	0.03 ±0.014	0.20 ±0.008	0.00 ±0.000	0.27 ±0.036	-	-	-	-	0.19 ±0.010	0.30 ±0.026	0.00 ±0.000	0.70 ±0.007
(D) Two competing	0.48 ±0.006	0.36 ±0.008	0.52 ±0.015	0.54 ±0.016	-	-	-	-	0.73 ±0.018	0.54 ±0.022	0.47 ±0.030	0.85 ±0.019
(E) Many answers												
(E1) Non correct	0.01 ±0.004	0.00 ±0.000	0.00 ±0.000	0.00 ±0.000	0.01 ±0.010	0.00 ±0.000	0.00 ±0.000	0.00 ±0.000	-	-	-	-
(E2) Correct appears	0.38 ±0.009	0.21 ±0.006	0.42 ±0.015	0.38 ±0.009	0.09 ±0.010	0.17 ±0.034	0.36 ±0.020	0.62 ±0.035	-	-	-	-
	Llama-8b											
	TriviaQA				**Math**				**Winobias**			
Error type	Greedy	Sampling	Majority	Probing	Greedy	Sampling	Majority	Probing	Greedy	Sampling	Majority	Probing
All	0.66 ±0.002	0.58 ±0.003	**0.68** ±0.003	**0.68** ±0.002	0.30 ±0.023	0.47 ±0.022	0.62 ±0.014	0.70 ±0.021	0.73 ±0.011	0.61 ±0.005	0.66 ±0.016	**0.84** ±0.006
(A) Refuses to answer	0.08 ±0.005	0.07 ±0.011	0.00 ±0.000	**0.16** ±0.011	0.00 ±0.007	0.04 ±0.015	0.00 ±0.000	0.25 ±0.025	-	-	-	-
(B) Consistently correct												
(B1) All	1.00 ±0.000	1.00 ±0.000	1.00 ±0.000	1.00 ±0.000	-	-	-	-	-	-	-	-
(B2) Most	0.98 ±0.001	0.87 ±0.002	**1.00** ±0.000	0.95 ±0.002	0.77 ±0.024	0.88 ±0.025	1.00 ±0.000	0.97 ±0.014	0.98 ±0.005	0.75 ±0.004	**1.00** ±0.000	0.94 ±0.003
(C) Consistently incorrect												
(C1) All	0.00 ±0.000	0.00 ±0.000	0.00 ±0.000	0.00 ±0.000	-	-	-	-	-	-	-	-
(C2) Most	0.06 ±0.013	0.18 ±0.009	0.00 ±0.000	**0.35** ±0.043	-	-	-	-	0.25 ±0.026	0.29 ±0.023	0.00 ±0.000	**0.65** ±0.022
(D) Two competing	0.44 ±0.029	0.42 ±0.035	0.53 ±0.020	**0.66** ±0.030	-	-	-	-	0.73 ±0.025	0.47 ±0.019	0.41 ±0.037	**0.86** ±0.014
(E) Many answers												
(E1) Non correct	0.00 ±0.000	0.00 ±0.000	0.00 ±0.000	0.00 ±0.000	0.00 ±0.000	0.00 ±0.000	0.00 ±0.000	0.00 ±0.000	-	-	-	-
(E2) Correct appears	0.46 ±0.009	0.34 ±0.009	0.53 ±0.007	**0.54** ±0.005	0.14 ±0.015	0.17 ±0.025	0.44 ±0.047	0.65 ±0.031	-	-	-	-

Table 14: Various answer choice strategies, instruct models.

	Mistral-7b-Instruct											
	TriviaQA				Math				Winobias			
Error type	Greedy	Random	Majority	Probing	Greedy	Random	Majority	Probing	Greedy	Random	Majority	Probing
All	0.63 ±0.003	0.64 ±0.002	0.67 ±0.004	0.71 ±0.003	0.55 ±0.021	0.52 ±0.019	0.57 ±0.025	0.70 ±0.014	0.77 ±0.012	0.77 ±0.008	0.77 ±0.010	0.79 ±0.008
(A) Refuses to answer	0.06 ±0.005	0.06 ±0.011	0.00 ±0.000	0.28 ±0.009	-	-	-	-	-	-	-	-
(B1) All	1.00 ±0.000	1.00 ±0.000	1.00 ±0.000	1.00 ±0.000	1.00 ±0.000	1.00 ±0.000	1.00 ±0.000	1.00 ±0.000	1.00 ±0.000	1.00 ±0.000	1.00 ±0.000	1.00 ±0.000
(B2) Most	0.88 ±0.007	0.83 ±0.009	0.99 ±0.002	0.89 ±0.010	0.87 ±0.013	0.84 ±0.024	1.00 ±0.000	0.96 ±0.007	0.91 ±0.031	0.87 ±0.029	0.96 ±0.017	0.89 ±0.032
(C) Consistently incorrect												
(C1) All	0.00 ±0.003	0.00 ±0.000	0.00 ±0.000	0.00 ±0.000	0.05 ±0.020	0.00 ±0.000	0.00 ±0.000	0.00 ±0.000	0.00 ±0.000	0.00 ±0.000	0.00 ±0.000	0.00 ±0.000
(C2) Most	0.11 ±0.009	0.15 ±0.012	0.00 ±0.000	0.53 ±0.005	0.10 ±0.040	0.20 ±0.050	0.00 ±0.000	0.82 ±0.037	0.18 ±0.057	0.20 ±0.039	0.00 ±0.000	0.54 ±0.067
(D) Two competing	0.32 ±0.010	0.45 ±0.023	0.50 ±0.024	0.78 ±0.017	-	-	-	-	-	-	-	-
(E) Many answers												
(E1) Non correct	0.01 ±0.003	0.00 ±0.000	0.00 ±0.000	0.00 ±0.000	-	-	-	-	-	-	-	-
(E2) Correct appears	0.23 ±0.020	0.19 ±0.022	0.38 ±0.009	0.56 ±0.025	-	-	-	-	-	-	-	-
	Llama-8b-Instruct											
	TriviaQA				Math				Winobias			
Error type	Greedy	Sampling	Majority	Probing	Greedy	Sampling	Majority	Probing	Greedy	Sampling	Majority	Probing
All	0.69 ±0.003	0.67 ±0.001	0.71 ±0.002	**0.73** ±0.004	0.89 ±0.010	0.87 ±0.012	**0.91** ±0.013	**0.91** ±0.010	0.75 ±0.009	0.74 ±0.009	0.76 ±0.012	**0.83** ±0.009
(A) Refuses to answer	0.06 ±0.011	0.05 ±0.011	0.00 ±0.000	**0.27** ±0.025	-	-	-	-	-	-	-	-
(B) Consistently correct												
(B1) All	1.00 ±0.000	1.00 ±0.000	1.00 ±0.000	1.00 ±0.000	1.00 ±0.000	1.00 ±0.000	1.00 ±0.000	1.00 ±0.000	1.00 ±0.000	1.00 ±0.000	1.00 ±0.000	1.00 ±0.000
(B2) Most	0.93 ±0.002	0.86 ±0.009	**1.00** ±0.001	0.92 ±0.004	0.94 ±0.014	0.92 ±0.014	**1.00** ±0.000	0.95 ±0.013	0.94 ±0.006	0.88 ±0.010	1.00 ±0.000	0.93 ±0.011
(C) Consistently incorrect												
(C1) All	0.00 ±0.001	0.00 ±0.000	0.00 ±0.000	0.00 ±0.000	-	-	-	-	0.00 ±0.000	0.00 ±0.000	0.00 ±0.000	0.00 ±0.000
(C2) Most	0.12 ±0.018	0.22 ±0.010	0.00 ±0.000	**0.43** ±0.010	-	-	-	-	0.11 ±0.018	0.15 ±0.025	0.00 ±0.000	**0.67** ±0.016
(D) Two competing	0.43 ±0.017	0.42 ±0.014	0.46 ±0.016	**0.60** ±0.010	-	-	-	-	0.39 ±0.068	0.39 ±0.047	0.38 ±0.042	**0.83** ±0.050
(E) Many answers												
(E1) Non correct	0.00 ±0.002	0.00 ±0.000	0.00 ±0.000	0.00 ±0.000	-	-	-	-	-	-	-	-
(E2) Correct appears	0.28 ±0.006	0.28 ±0.008	0.40 ±0.009	**0.52** ±0.009	-	-	-	-	-	-	-	-